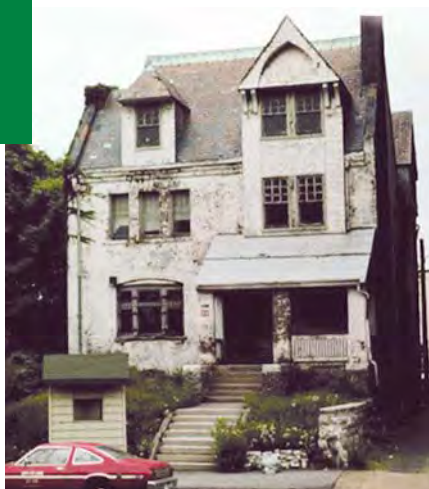


# EARTHWAYS CENTER TEACHES BY EXAMPLE

**This St. Louis, Missouri, teaching center promotes green building and home performance through its guided tours and training programs while serving as a model for green and energy-efficient design.**



EARTHWAYS ARCHIVES

(above) The urban house that became EarthWays Center had deteriorated to a crumbling point when it was selected for the complete renovation. (right) The original nineteenth century architectural features have been restored, providing a handsome historic context for twenty-first century demonstrations of sustainable building and lifestyle choices.

by **JEAN PONZI**



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**E**arthWays Center, a division of Missouri Botanical Garden, promotes sustainable energy and resource use. Based in a stately old brick home in midtown St. Louis that has been completely renovated to model sustainability in buildings and personal lifestyle choices, its educational programs have earned respect throughout Missouri and around the Midwest, addressing energy efficiency, recycling and waste reduction, air quality, and green building. Expertise of the current staff includes environmental education, public policy, engineering, media communications, and multiple specialties they like to call applied sustainability.

This 1885 vintage Victorian has been through cycles of major change. Originally a gracious single-family home, it was abandoned by its last live-in owner and stood vacant for over 13 years. The house had become a derelict shell by the early 1990s, an ideal candidate for a gut rehab. A local environmental group raised \$1½ million in cash and contributed goods and services, and redeveloped the site for educational use. It was opened to the public in 1994.

The EarthWays facility houses an impressive array of green features, including a groundsource heating and cooling system for the first floor and a high-efficiency gas furnace for the sec-

ond floor; 2 x 6 reframing of all exterior walls and superinsulation using blown-in cellulose made from recycled newsprint; high-efficiency windows; compact fluorescent lighting; efficient appliances; sustainable flooring; aerated and low-flow plumbing fixtures; a PV installation; and a variety of handsome reused and recycled-content furnishings. New features are regularly added to this mix.

## Whole-House Perspective

An educational challenge facing EarthWays today is advancing the integration of green building concepts with building performance practices, particu-

larly among homeowners. Reputable national evaluation tools that support this kind of work include the EPA Energy Star programs, the U.S. Green Building Council (USGBC) Leadership in Energy and Environmental Design (LEED) rating systems, and contractor certification programs like those provided by the Building Performance Institute.

"Of course we still teach about fundamentals like energy efficiency and recycling in the home environment," says EarthWays' Program Manager Glenda Abney.

"As market acceptance grows for specific building products and practices that are verifiably green, we are challenged to communicate how materials, building systems, and everyday operations and maintenance have to work together to make living environments healthy, safe, and comfortable as well as efficient."

As environmental educators, the EarthWays team compares home per-

formance to cycles and flows in the natural world, where everything is connected to everything else. Modifying one thing will affect another. It's true for homes and ecosystems. If we fix what we think is not working, without thinking through the problem to find its real source, the fix will probably fail—and the problem may get worse.

"Guided tours of EarthWays Center have been a good way for us to educate visitors of all ages about the green building practices and products they see here," explains Abney. "We can introduce our visitors to concepts of whole-building science as we take them through the basic EarthWays tour, where they see a positive flow of interconnected products and systems. Showing them our cellulose insulation through kitchen wall cutaways and then explaining how a ground source system works, as they view the renewable energy murals painted on the walls in our garden, can lead to discussions of the crit-

ical interaction between good insulation, properly sized HVAC units, and effective air sealing. Tours also highlight the interrelationships behind our solar-powered kitchen."

EarthWays' appliances and lighting can be run on a mere 1.5 kW PV array because compact fluorescent bulbs are installed in all the fixtures and there is a high-efficiency SunFrost refrigerator. "The refrigerator alone," notes Abney, "models the importance of insulation and designing for efficiency." The SunFrost's exterior walls and space between compartments incorporate 3 to 5 inch insulated spaces. Separate, highly efficient compressors maintain freezer and refrigerator section temperatures. The SunFrost's compressors are mounted on top of the appliance, compared to designs that stow these knob-

## Home Performance Flourishing in St. Louis

Home Performance with Energy Star, a program of EPA and DOE, has partner organizations in a dozen cities around the country. These partner organizations manage the program locally, with operating budgets drawn from various sources, including public benefits funds, state budgets, and utility company support. Smaller programs, such as the ones in Missouri, are run on a combination of support from federal and state energy budgets, in-kind contributions, local contractor support, and creative work by the nonprofits that run the programs.

The St. Louis Home Performance program started in 2004, one year after the program was initiated in Kansas City. Both efforts are maintained largely through grants from the Missouri Department of Natural Resources (MO DNR). Missouri Botanical Garden's EarthWays Center is the coordinating partner for the St. Louis metro area program. EarthWays Center staff works with the Building Performance Institute, a source of independent third-party quality control, to provide certification and accreditation

for participating contractors and their companies.

EarthWays' mobilization of the program has involved a committed, person-to-person approach, driven by a big vision of what Home Performance can achieve. From the first round of recruitment efforts, 23 contractors requested information on the program, 9 have been trained, and 6 are formally engaged in the program. There is a waiting list for the next cycle of training. In the first six months, EarthWays staff took over 40 requests for referrals that resulted in 17 whole-building audits with five of these clients choosing to make physical improvements in their homes. Each household that receives a whole building audit and subsequent home improvements proves the worth of continuing these efforts, and provides a local case study with significant market value.

Marketing the principles behind Home Performance with Energy Star is critical to advancing this work. EarthWays Center is strategically positioned to reach receptive homeowners, working in a unique green home demonstration facility and being

widely known as a resource for public education in this kind of subject matter, including themed events like the EarthWays Energy Festival.

Participation statistics to date may seem small, but when very limited budgets and the program's start-up status are factored in, staff is confident that the early results are a preview of what can be achieved with additional time and marketing.

Home Performance with Energy Star is a great vehicle to support EarthWays' goal of combining residential green building techniques with whole-building performance. Supportive partners in this effort including the EPA, DOE, MO DNR, Missouri Botanical Garden, the Building Performance Institute, the Kansas Building Science Institute, MSI Consulting, and UMB Bank, which offers financial packages for homeowners and contractors.

*To learn more about the St. Louis Home Performance with Energy Star program, contact Glenda Abney at (314) 577-0288 or visit [www.earthwayscenter.org](http://www.earthwayscenter.org).*

shaped components under conventional models, letting waste heat rise through cooled spaces.

### Sleuthing Solutions

Through a grant coordinated by the Missouri Department of Natural Resources, EarthWays Center has established a home performance program for the greater St. Louis area (see “Home Performance Flourishing in St. Louis”). Local contractors were trained at EarthWays Center, using its features as a learning lab. Frequent public events, such as the EarthWays Energy Festival, provide an ideal venue to market home performance concepts and services to a receptive audience of

homeowners. The certified contractors are providing whole-building audits for St. Louis homeowners, and effective home improvements have begun.

“When people come to EarthWays Center now,” says Abney, who also coordinates the St. Louis Home Performance program, “we can educate them much more effectively about the possibilities. Working with the contractors in the home performance program, homeowners can turn that education into reality.”

Results of investigations by the St. Louis cadre of home performance contractors give the EarthWays team intriguing stories to use in promoting the program. For example, one family thought they had a major lint problem—that somehow their clothes dryer was blowing lint all over the house on a daily basis. Their home performance contractor discovered an opening in the duct system in the attic and improper insulation that was being pulled into that hole. The insulation was being carried through the duct system and blown all over the house. Duct sealing, not a new dryer, was needed to contain the “lint.”

Another woman reported a paint problem. After two contractors repainted, she still had peeling paint. Her home performance contractor found that she also had air leaks in that area of her house, causing condensation behind the walls that made the paint peel continually. After an air sealing job was completed, only one more painter got a job at that home.

Then there was the family who had high utility bills. Working with a home performance contractor, they discovered that they had no insulation in the walls or attic. In addition, they had a severe mold issue. Getting a bigger air conditioner or simply cleaning up the mold wouldn’t make the problem go away. They had to fix a combination of water, insulation, and ventilation problems.

“Home Performance with Energy Star supports our work with homeowners like these to develop understanding of how a house functions as a whole system,” says Abney. “The St. Louis home performance contractors corrected all of these problems, but if these people had not consulted with a building scientist, they would have seen only part of the issue and would indeed still have their problems.”

“When people call on us with problems, we can confidently refer them to certified professionals who can evaluate the components of their home’s system, and then implement improvements as an owner-contractor team. And we can direct people to financial support for this process through UMB Bank’s low-interest lending partnership in our Home Performance program.”

### A Wealth of Potential Benefits

The EarthWays Center team takes dozens of calls a week from people seeking help for problems in their homes, as well as those looking to build new or modify existing homes. Visitors to EarthWays events tend to spend time exploring and asking in-depth questions. Many of these people are looking for maximum benefits: comfort, durability, safety, health, and the most effective use of natural resources. Staff responses promote a combination of green building techniques and home performance principles. Says Abney, “Our goal is to see housing stock built and remodeled in this region using green building products and methods and operated as effective whole building systems. And this combination will achieve our ultimate goal, which is to conserve natural resources.”

Contractors involved with the EarthWays Center Home Performance program are also seeking maximum benefits, and they are willing to invest time and resources to achieve them—for their

own businesses and for their clients. Ed Fieser, owner of MaxHome Performance Contracting, was one of the first St. Louis contractors to earn home performance certification. “I was tired of working for big businesses where people didn’t count,” he reports. “I wanted to find something satisfying and beneficial to the people I was working for. Home performance does that for me, and it helps everyone. When I go home at the end of the day, I feel good because what I’ve done benefits the person I’m working for, benefits me financially, and benefits my family. Really, everyone around me benefits, because the efficiencies I’ve increased will improve the natural world by reducing emissions. I’ve harmed no one and helped everyone, so I’m always assisting people, no matter what I’m dealing with.”

### A Hub of Green Activity

In addition to its pivotal role in Home Performance with Energy Star, EarthWays Center is also home to operations of the USGBC’s St. Louis Regional Chapter (“Green Me in St. Louis”). Deborah Chollet Frank, director of EarthWays Center, helped to establish the chapter in 2001, and she serves as chapter coordinator. She and her staff have persisted through the hard-won evolution of these movements. “There were several earlier attempts to launch a commercial green building group in St. Louis,” recalls Frank, “made by people who were ardent advocates. But it took a while to reach the critical mass of commitment that secured a chapter charter from USGBC.”

“Although chapter activity has been primarily focused in the commercial sector so far,” adds Frank, “we get many inquiries from people who want to incorporate green factors into their homes, who are looking for professional resources to do so. There is strong potential here for a green homes movement.”

Green building in the St. Louis residential sector is gaining momentum. A handful of local builders market green as their specialty, and some are actively supporting development and rollout of USGBC’s LEED for Homes. The Home Builders Association of Greater St. Louis rallied national and regional support for a fall 2005 launch of their own green building program.



Matt Belcher, owner of Belcher Homes in St. Louis, is a longtime advocate of green building. He helped his contractor father build their family home in 1960, incorporating resource-efficient methods and materials, which, he notes, “are easier now with better technologies behind the sustainable products.”

“You have to think about all the systems that go into a house, and how they will interact,” says Belcher, “so all of them work together as efficiently as possible. You also have to take the

lifestyle of the customers into consideration, and educate them to get the most out of the features of their home. It’s really a superior way of using the process of building.”

Jim Holtzman, principal of Miracle Design/Build, has been committed to green building since college projects involved him with solar home design and construction. He currently partners on high-end St. Louis custom home projects, through Evolution Developments LLC, with builder Jay Jarvis,

owner of RESAC Corporation. Says Holtzman, “These houses maximize energy efficiency, air flow, and daylight. Open floor plans let people circulate freely through their living spaces, without barriers or closed-in hallways. Homes are oriented on their sites to take advantage of natural climate control factors and connect indoor and outdoor spaces. The quality factor of green is evident in materials and construction, and

## Green Me in St. Louis

The EarthWays Center is in many ways the hub of a vital, expanding green building movement in eastern Missouri. One of the many green building initiatives that EarthWays supports is the USGBC St. Louis Regional Chapter. The chapter, which now boasts 250 members representing every segment of the building industry, was founded in 2001 with an ambitious mission: to transform the built environment in a metropolitan market comprised of urban, suburban, and rural areas of Missouri and Illinois. Volunteer committees pursue this goal through strategic work with local government and business leaders, embodying the USGBC slogan “Build Green—Everyone Profits.”

Deborah Chollet Frank oversees this partnership as a Missouri Botanical Garden division director and the USGBC St. Louis Regional Chapter coordinator. She cites the Garden’s institutional credibility—including the international influence of Garden Director Peter H. Raven, a tireless advocate of sustainability—as a factor in the vitality of green building activity in the St. Louis metro area. Says Frank, “This relationship is a practical asset to both organizations. With access to Garden facilities, we can host chapter events in beautiful, accessible settings that affirm connections between our work with buildings and the natural world. The Green Building Council and EarthWays Center cross-reference resources for commercial and residential audiences. Outreach power is multiplied, enabling



DEBORAH CHOLLET FRANK

The U.S. Green Building Council St. Louis Regional Chapter sponsors monthly programs that often feature tours of the area’s growing number of LEED-certified green buildings.

all partners to better provide information and services.”

As do all USGBC chapters, the St. Louis chapter measures its accomplishments using USGBC’s international standard, the LEED rating systems. The regional chapter set and is rapidly closing in on, a major goal: having 30 metro area buildings registered for or holding LEED certification by the end of 2006. Over 20 public and private projects are now using LEED criteria in new commercial construction, renovation and operation of existing buildings, and commercial interior design. Supporting this movement, over 250 individuals working in the region hold LEED accreditation.

Education is a priority for USGBC-St. Louis. Monthly programs combine prime networking opportunities with presentations that detail elements of building green, continually attracting new recruits. Frequent professional trainings are geared for designers, builders, engineers, owners, and managers. Tours of sites where LEED is applied draw capacity crowds, and an annual Green Product Expo attracts regional and national vendors of innovative wares.

*Visit [www.usgbc.org](http://www.usgbc.org) to learn more about USGBC, LEED, and the St. Louis Regional Chapter.*

most importantly, in the feel of the living environment.”

### Making the Case for Integrating Green and Performance

“EarthWays Center staff members are personally motivated by a desire to protect natural resources,” says Glenda Abney, “but we know we have to make a business case for the principles we preach. Our region’s economics are so different from those that drive successful greening efforts on both coasts. You could say we are both blessed and cursed with relatively low utility rates, waste disposal costs, and other dollar factors that elsewhere boost payback on performance and efficiency investments.”

In Missouri, little political capital has been invested in promoting green building and home performance initiatives. Fiscal incentives available in other parts of the country, based in utility deregulation and tax abatements, are not in play in the Show-Me State, although communities on the Illinois side of the St. Louis metro area do have some of these advantages.

Given these conditions, Abney reminds people to consider home performance as a form of home health insurance. “Many people understand how wellness practices keep personal quality of life high and health care costs to a minimum. Home performance offers this principle of prevention for our biggest financial investments, our homes. Homeowners who are attracted to green building have already made connections between environmental concerns, health, economics, and the fact that their choices have some power. This audience is primed to appreciate what home performance has to offer.”

### Leadership Through Education

EarthWays Center has always taught about systems and cycles in the natural world. In the K-12 programs that the organization originally focused on, students developed energy and recycling projects and experienced how these human efforts mimic the effectiveness of natural interactions, such as solar power



The annual EarthWays Energy Festival features exhibits, demonstrations, and the chance to interact with green product vendors and home performance experts.

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in the food chain and decomposition. In current green building partnerships, EarthWays tries to get all players in the building process—owners, designers, engineers, contractors, auditors—to work together from the earliest planning discussions, functioning like an ecosystem, rather than passing a project between their isolated silos.

“As environmental educators, we strive to provide children and teachers with hands-on experience of natural systems and how systems thinking clearly beats linear logic,” says Abney. “We also have to teach our professional partners and the public how the buildings we live in are like the interconnected systems in the natural world. The parts have to function as a whole to make effective use of all their green features. And the results will be increased comfort and safety, energy savings, improved indoor air quality, and conservation of water and other natural resources. If planning is sound, dollars will be saved as well.”

Adds Frank, “The St. Louis commercial building industry has begun to accept and value these concepts. We want to make sure our area’s residential building industry is heading in that same direction.”

Part of the process is sharing stories that show why others get involved. The St. Louis region’s initial leaders in green building and home performance have mostly stepped up because they believe

that investing in green building is the right thing to do. As their projects proliferate and local data circulate, others are seeing business and personal advantages in these practices. Education is prompting more service providers and more customers to get involved.

EarthWays’ public education projects in the planning stages include establishing a national database and resource-sharing network of demonstration sites like EarthWays Center, and taking the team’s educational message out to the huge audiences that attend home shows in St. Louis. These are exciting new opportunities to provide outreach about integrating green building with home performance.

*Jean Ponzi is a program manager at EarthWays Center.*

### For more information:

EarthWays Center is located at 3617 Grandel Square, in midtown St. Louis. Public tours are offered the third weekend of each month, and organized group tours may be scheduled with two weeks’ advance notice. Learn more at [www.earthwayscenter.org](http://www.earthwayscenter.org), or by calling (314)577-0220.